Econometric_HW0331

#4.1

4.1 Answer each of the following:

a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2.

b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.

c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2.

a. $R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} = 0.8429$ *

$SSTO = 7825.94 - 19.21^2 \cdot 20 = 445.458$

b. $R^2 = \frac{SSR}{725.94} = 0.7911 \Rightarrow SSR = 574.2911$

$SSE = 725.94 - 574.2911 = 151.6489$

$MSE = 151.6489 \div (20-2) = 8.4249 = \sigma^2$

c. $SSR = 631.63 - 182.85 = 448.78$

$R^2 = 448.78 / 631.63 = 0.7105$ *

#4.3

4.3 We have five observations on x and y. They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^{5} \hat{e}_i^2 = 3.6$, $\sum_{i=1}^{5} (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^{5} (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

a. the predicted value of y for $x_0 = 4$.
b. the $\text{se}(f)$ corresponding to part (a).
c. a 95% prediction interval for y given $x_0 = 4$.
d. a 99% prediction interval for y given $x_0 = 4$.
e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a. $\hat{y}_i = 1.2 + 0.8 \cdot 4 = 4.4$

b. $se(f) = \sqrt{\widehat{var}(f)} = \sqrt{\hat{\sigma}^2\left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}\right]} = \sqrt{1.2\left[1 + \frac{1}{5} + \frac{(4-1)^2}{10}\right]} = \sqrt{2.52} \doteq 1.5875$

$\hat{\sigma}^2 = MSE = 3.6 \div (5-2) = 1.2$

$\bar{x} = (3+2+1-1+0) \div 5 = 1$

c. $95\%\ P.I. = \left[\hat{y}_0 \pm t_{0.025}(3) \cdot se(f)\right] = \left[4.4 \pm 3.182 \cdot 1.5875\right] = [-0.6514, 9.4514]$ ※

d. $99\%\ P.I. = \left[\hat{y}_0 \pm t_{0.005}(3)\ se(f)\right] = \left[4.4 \pm 5.841 \cdot 1.5875\right] = [-4.8726, 13.6726]$ ※

e. $95\%\ PI = \left[2 \pm t_{0.025}(3) \cdot 1.2\right] = [-1.8184, 5.8184]$ ※

$se(f) = \sqrt{1.2\left[1 + \frac{1}{5} + \frac{(1-1)^2}{10}\right]} = \sqrt{1.44} = 1.2$

$\hat{y}_0 = 1.2 + 0.8 \cdot 1 = 2$

→ The prediction interval is narrowest at the sample mean $x_0 = \bar{x}$ because the term $(x_0 - \bar{x})^2$ is minimized.

As we move away from the mean, the interval become wider.

#4.25(a)

	Dependent variable:
	log(price)
sqft	0.036***
	(0.001)
Constant	4.394***
	(0.043)
Observations	500
R2	0.542
Adjusted R2	0.541
Residual Std. Error	0.340 (df = 498)
F Statistic	588.686*** (df = 1; 498)
Note:	*p<0.1; **p<0.05; ***p<0.01

	Item	Value
1	slope	9.020
2	elasticity	0.983

The estimated regression is $\ln\widehat{(PRICE)}_i = 4.394 + 0.036\widehat{SQFT}$.

$\widehat{\beta_1}$=4.394→When $\widehat{SQFT}$=0,the predicted value of $ln\widehat{(PRICE)}_i$ is 4.394. This represents the nature logarithm of price for a house with zero square feet.

$\widehat{\beta_2}$=0.036→The coefficient is statistically significant, and this means that house price will change 100*0.036=3.6% for every one-unit increase in square feet.

#4.25(b)

	Dependent variable:
	log(price)
log(sqft)	1.025***
	(0.048)
Constant	2.050***
	(0.158)
Observations	500
R2	0.474
Adjusted R2	0.473
Residual Std. Error	0.364 (df = 498)
F Statistic	448.488*** (df = 1; 498)
Note:	*p<0.1; **p<0.05; ***p<0.01

	Item	Value
1	slope	9.400
2	elasticity	1.025

The estimated regression is $\ln\widehat{(PRICE)}_i = 2.050 + 1.025\ln\widehat{(SQFT)}$.

$\widehat{\beta_1}$=2.050→When $\ln\widehat{(SQFT)}$=0,the predicted value of $ln\widehat{(PRICE)}_i$ is 2.050. This represents the nature logarithm of price for a house with 1 square feet.

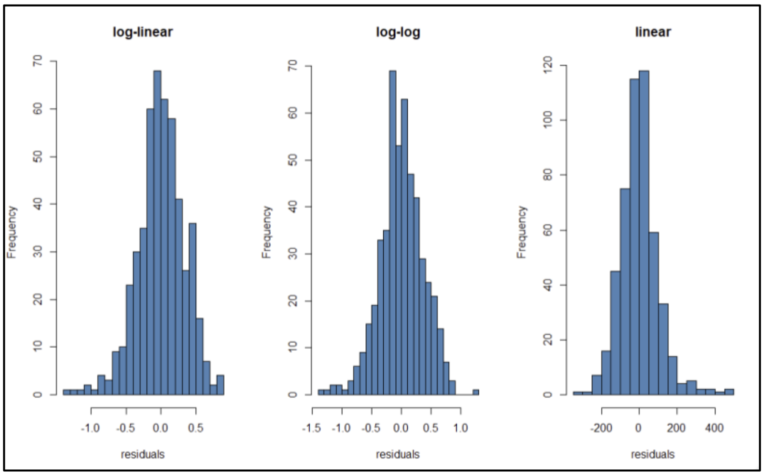
$\widehat{\beta_2}$=1.025→The coefficient is statistically significant, and this means that house price will change 100*1.025=102.5% for every 1% increase in square feet.

#4.25(c)

	Item	Value
1	generalized R2_a	0.662
2	generalized R2_b	0.645
3	R2_c	0.641

The higher generalized R^2 for the non-linear models indicates that they provide a superior fit to the house price data and offer greater explanatory power than the standard linear specification.

#4.25(d)

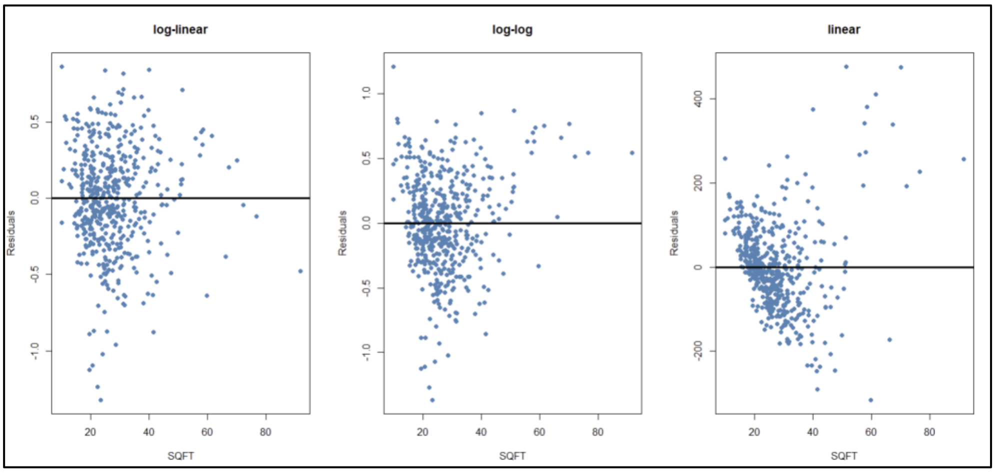


Jarque-Bera Test Results

	Model	X_squared	df	p_value
1	residuals_a	26.6790	2	0.000002
2	residuals_b	14.2787	2	0.0008
3	residuals_c	221.1115	2	0

Based on the Jarque-Bera test results, all three models have p-values significantly lower than the 0.05 threshold ($p < 0.001$ for all). Therefore, we reject the null hypothesis of normality in every case. The histograms also show visible skewness and heavy tails, indicating that the distributions of the residuals are not compatible with the assumption of normality.

#4.25(e)



The residual plots show some differences across the three models. In the linear model, the residuals display a clear pattern and increasing spread as SQFT increases, suggesting heteroskedasticity and possible model misspecification.

In contrast, the log-linear and log-log models show residuals that are more evenly scattered around zero, with less obvious patterns. The variance of the residuals appears more stable across different values of SQFT in these models.

Overall, the log specifications (log-linear and log-log) seem to provide a better fit than the linear model, as their residuals are more randomly distributed and closer to satisfying the classical regression assumptions.

#4.25(f)

	Model	Predict_Value
1	A	214.234
2	B	227.539
3	C	246.456

#4.25(g)

	Model	Fit	Lower	Upper
1	A	214.234	109.768	418.117
2	B	227.539	111.141	465.841
3	C	246.456	44.277	448.634

#4.25(h)

Based on the comprehensive analysis, the Log-Log model (Model B) is the most appropriate functional form for this dataset.

Reasons:

- **Superior Fit:** The Log-Log model yields a higher generalized R^2 compared to the linear model, indicating it explains a larger proportion of the variation in house prices.
- **Normality of Residuals:** According to the Jarque-Bera test, although all models technically reject the null hypothesis of normality, the Log-Log model has the lowest X^2 statistic (14.27), suggesting its residuals are the closest to being normally distributed.
- **Homoskedasticity:** The residual plots show that the Log-Log transformation successfully stabilized the variance. Unlike the linear model, which exhibits clear heteroskedasticity (a funnel shape), the residuals of the Log-Log model are randomly scattered around zero.

#5.5

5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T\to\infty} \text{var}(b_2|\mathbf{x}) = 0$.

a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \ldots, T$. Note that $\mathbf{x} = (1, 2, \ldots, T)$, $\sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2 / T$, $\sum_{t=1}^{T} t = T(T+1)/2$ and $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$.

b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \ldots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \ldots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \cdots + r^n = \frac{r(1-r^n)}{1-r}$$

c. Provide an intuitive explanation for these results.

$$Var(b_2|x) = Var\left(\frac{\Sigma(X_t-\bar{X})Y_t}{\Sigma(X_t-\bar{X})^2}\right) = \frac{\Sigma(X_t-\bar{X})^2}{\Sigma(X_t-\bar{X})^4} var(Y_t) = \frac{\sigma^2}{\sum_{t=1}^{T}(X_t-\bar{X})^2}$$

a. $$\because X_t = t, \ \sum_{t=1}^{T}(X_t-\bar{X})^2 = \sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \frac{(\Sigma t)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{\left(\frac{T(T+1)}{2}\right)^2}{T}$$

$$= \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} = \frac{T^3-T}{12}$$

$$\lim_{T\to\infty} Var(b_2|X) = \lim_{T\to\infty} \frac{12\sigma^2}{T^3-T} = 0 \rightarrow \text{Consistent}$$

b. $$\sum_{t=1}^{T} X_t = 0.5 + 0.5^2 + \cdots + 0.5^T = \frac{0.5(1-0.5^T)}{1-0.5} = 1-0.5^T$$

$$\sum_{t=1}^{T} X_t^2 = 0.5^2 + (0.5^2)^2 + \cdots + (0.5^T)^2 = \frac{0.5^2(1-0.5^{2T})}{1-0.5^2} = \frac{1-0.5^{2T}}{3}$$

$$\Sigma(X_t-\bar{X})^2 = \Sigma X_t^2 - \frac{(\Sigma X_t)^2}{T} = \frac{1-0.5^{2T}}{3} - \frac{(1-0.5^T)^2}{T}$$

$$\lim_{T\to\infty} \Sigma(X_t-\bar{X})^2 = \lim_{T\to\infty} \frac{1-0.5^{2T}}{3} - \frac{(1-0.5^T)^2}{T} = \frac{1-0}{3} - \frac{(1-0)^2}{\infty} = \frac{1}{3}$$

$$\Rightarrow \lim_{T\to\infty} Var(b_2|X) = \lim_{T\to\infty} \frac{\sigma^2}{1/3} = 3\sigma^2 \neq 0 \rightarrow \text{not Consistent}$$

c. Model (a) is consistent because the regressor x_t = t grows indefinitely, providing "infinite information" as the sample size increases, which drives the estimation error to zero. In contrast, Model (b) is inconsistent because x_t = 0.5^t decays toward zero so rapidly that new data points eventually provide no additional information, causing the total variation Sxx to stay finite and the estimation error to never vanish.

#5.6

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad \widehat{\operatorname{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

a.

① $\begin{cases} H_0: \beta_2 = 0 \\ H_a: \beta_2 \neq 0 \end{cases}$ ② $\alpha = 0.05$ ③ $T = \dfrac{\hat{\beta}_2 - 0}{SE(\hat{\beta}_2)} \overset{H_0}{\sim} t(63-3)$

④ $RR = \{ |T| \geq t_{0.025}(60) = 2 \}$ ⑤ $T_0 = \dfrac{3-0}{\sqrt{4}} = 1.5$

⑥ ∵ $T_0 \notin RR$，故 do not reject H_0

b. ① $\begin{cases} H_0: \beta_1 + 2\beta_2 = 5 \\ H_a: \beta_1 + 2\beta_2 \neq 5 \end{cases}$ ② $\alpha = 0.05$ ③ $T = \dfrac{\hat{\beta}_1 + 2\hat{\beta}_2 - 5}{SE(\hat{\beta}_1 + 2\hat{\beta}_2)} \overset{H_0}{\sim} t(63-3)$

$SE(\hat{\beta}_1 + 2\hat{\beta}_2) = \sqrt{3 + 4 \cdot 4 + 4 \cdot (-2)} = \sqrt{11}$

④ $RR = \{ |T| \geq t_{0.025}(60) = 2 \}$ ⑤ $T_0 = \dfrac{2 + 2 \cdot 3 - 5}{\sqrt{11}} = 0.9045$

⑥ ∵ $T_0 \notin RR$，故 do not reject H_0 #

c. ① $\begin{cases} H_0: \beta_1 - \beta_2 + \beta_3 = 4 \\ H_a: \beta_1 - \beta_2 + \beta_3 \neq 4 \end{cases}$ ② $\alpha = 0.05$ ③ $T_0 = \dfrac{\hat{\beta}_1 - \hat{\beta}_2 + \hat{\beta}_3 - 4}{SE(\hat{\beta}_1 - \hat{\beta}_2 + \hat{\beta}_3)} \overset{H_0}{\sim} t(63-3)$

$SE(\hat{\beta}_1 - \hat{\beta}_2 + \hat{\beta}_3) = \sqrt{3 + 4 + 3 - 2(-2) + 2} = \sqrt{16} = 4$

④ $RR = \{ |T| \geq t_{0.025}(60) = 2 \}$ ⑤ $T_0 = \dfrac{2 - 3 - 1 - 4}{4} = -1.5$

⑥ ∵ $T_0 \notin RR$，故 do not reject H_0 #

#5.20

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \ldots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means (for the first and second halves) of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2,mean}$ is unbiased.

b. Derive an expression for $\text{var}(\hat{\beta}_{2,mean}|\mathbf{x})$.

c. Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.

d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2?

a. $\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \frac{\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 - \beta_1 - \beta_2 \bar{x}_1 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$

$\Rightarrow E(\hat{\beta}_{2,mean}) = \beta_2 + \frac{\overbrace{E(\bar{e}_2)}^{=0} - \overbrace{E(\bar{e}_1)}^{=0}}{\bar{x}_2 - \bar{x}_1} = \beta_2$ ※ [UE]

b. $Var(\hat{\beta}_{2,mean}|x) = Var\left(\beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}\right) = Var(\beta_2) + \frac{Var(\bar{e}_2) + Var(\bar{e}_1)}{(\bar{x}_2 - \bar{x}_1)^2}$

$= \frac{\sigma^2/N/2 + \sigma^2/N/2}{(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$ ※

c. $Var(\hat{\beta}_{2,mean}) = E[var(\hat{\beta}_{2,mean}|x)] + Var[\underbrace{E(\hat{\beta}_{2,mean}|x)}_{\beta_2}]$

$= E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right]$ ※

$4\ var(\beta_2) = 0$

d. $\lim_{N\to\infty} var(\hat{\beta}_{2,mean}|x) = \lim_{N\to\infty} \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = 0 \rightarrow$ consistent ※

$\Rightarrow$ when $N \to \infty$ ※

#5.20(e)(f)

	N	var_b2	var_b2_hat_mean	ex_1_sx2	ex_var
1	100	1.225	0.241	0.120	0.024
2	500	1.652	0.322	0.161	0.032
3	1000	0.122	0.120	0.120	0.120
4	5000	0.165	0.161	0.161	0.160

As the sample size increases, both variances decrease toward zero, indicating that both estimators are consistent. However, the OLS estimator consistently has a smaller variance than the mean-based estimator, showing that it is more efficient. The simulation results also confirm that $E[(s_x^2)^{-1}]$converges to 0.12 and $E[4/(\bar{x}_2-\bar{x}_1)^2]$converges to 0.16.

#5.20(g)(h)

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
   i. $\text{var}(b_2|\mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean}|\mathbf{x})$ where b_2 is the OLS estimator.
   ii. Estimates for $E\left[(s_x^2)^{-1}\right]$ and $E\left[4/(\bar{x}_2-\bar{x}_1)^2\right]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?

g. Show that $E\left[(s_x^2)^{-1}\right] \xrightarrow{p} 0.12$ and $E\left[4/(\bar{x}_2-\bar{x}_1)^2\right] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b-a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

g. $\sigma_X^2 = \dfrac{(10-0)^2}{12} = \dfrac{100}{12}$

① $E[(S_X^2)^{-1}] \xrightarrow{p} \dfrac{1}{\sigma_X^2} = \dfrac{12}{100} = 0.12$

② 前半部: $X_1, X_2, \cdots, X_{N/2} \sim U(0,5) \rightarrow \bar{X}_1 = 2.5$

後半部: $X_{N/2+1}, \cdots, X_N \sim U(5,10) \rightarrow \bar{X}_2 = 7.5$

$\Rightarrow \bar{X}_2 - \bar{X}_1 = 5$

$E\left[\dfrac{4}{(\bar{X}_2-\bar{X}_1)^2}\right] \xrightarrow{p} \dfrac{4}{5^2} = 0.16$ ※

h. If the observations are not ordered, the two subsample means become similar, making the denominator close to zero. This causes the variance to explode, so the estimator is not consistent.

e.g. $\bar{X}_1 \approx \bar{X}_2$, $\bar{X}_2 - \bar{X}_1 \rightarrow 0 \Rightarrow Var(\hat{\beta}_{2,mean}) \rightarrow \infty$, not consistent ※